Alfie R. Pelicano

Full-Stack Developer

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Professional Summary

Research Specialist and Geodetic Software Developer with 16+ years in scientific computing and spatial data. Working at the intersection of earth science and software engineering: processing GNSS and geophysical data for earthquake research while building the software to automate what shouldn't still be manual. Specialized in headless processing pipelines, distributed task systems, and full-stack applications — prioritizing minimalist, low-cognitive-load architectures.

Currently maintaining flagship earth science systems (MOVE Faults, EIL Hazard Suite), civic tech platforms (YAKAP-Link), and B2B marketplaces; development process is specification-driven and test-backed. **Core competencies:** GNSS Processing, Spatial Geohazards, Full-Stack Web Development, Agentic AI Workflows, Distributed Task Queues (Celery/Redis), Relational Databases (PostgreSQL).

Technical Skills

Frontend	Next.js 14 (App Router), React 19 (Server Components), Astro (Static Site Generation), TypeScript, Vanilla JavaScript, Tailwind CSS, shadcn/ui
Backend	Python / FastAPI, Node.js / Express, Celery (Distributed Task Queues), Redis, Docker, Agentic AI Workflows (LangGraph / Ollama), Prisma ORM, NextAuth.js
Database & Infra	PostgreSQL (Advanced SQL), Neon (Serverless), Supabase, Row Level Security (RLS)
Python & Scientific	Python 3.11+, Click CLI, Rich Terminal, Pandas, NumPy, SciPy, Shell Scripting, Fortran (Legacy Scientific Code)
Domain Expertise	GNSS Processing & Field Ops, Spatial Geohazards (Landslide / Seismic), Scientific Computing & HPC, Heuristic Engines, Constraint Solving
Testing & Quality	Spec-Driven Architecture & TDD, Playwright (E2E), Vitest (Unit), Pytest, Git Workflows
Tools & Workflows	Agentic Workflow Design & Orchestration, LaTeX / Beamer, ADR-driven Architecture, Linux / Bash

Production Applications

2024–Present **PipetGo – B2B Laboratory Testing Marketplace**, *Next.js 14 · TypeScript · Prisma · NextAuth · PostgreSQL*

Production MVP connecting businesses with ISO 17025 accredited testing labs.

- Implements multi-tiered pricing modes (QUOTE_REQUIRED, FIXED, HYBRID) and multi-role authorization workflows.
- Undergoing User Acceptance Testing with initial partner organizations before pilot deployment.
- 233 passing tests (Vitest + Playwright E2E) validating secure data access and quotation workflows.

Repository: `https://github.com/alfieprojectsdev/pipetgo`

2024–Present **PHAST – PHIVOLCS Hazard Assessment Tool**, *Vanilla JS · Chrome Extension API · Leaflet.js*

Chrome Extension command center that injects professional GIS controls into standard web maps.

- Dynamically generates formatted, official hazard assessment text directly from map interactions.
- Replaces manual text authoring for hazard assessors, significantly accelerating operational workflows.

Repository: `https://github.com/alfieprojectsdev/phat`

Active Development Projects

2024–Present **MOVE Faults & EIL Hazard Suite**, *Python · Celery · React · FastAPI · Docker*

Flagship earth science systems modernizing decades of legacy geodetic and hazard processing.

- **MOVE Faults:** GNSS automation monorepo featuring a Celery-backed distributed task queue, real-time NMEA TCP streaming, and an offline React field app.
- **EIL Hazard Suite:** Decoupled geohazard system with a headless spatial processing engine (EIL-Calc) for landslide susceptibility scoring and a React visualizer (EIL-Viz) for ground-truth validation.

2024–Present **Civic, Utility & Workflow Platforms**, *Next.js 14 · TypeScript · PostgreSQL · Prisma / Drizzle*

- **YAKAP-Link:** Civic tech proof-of-concept with strict transaction constraints and dispensing guard logic to mitigate phantom claims in the PhilHealth medicine program.
- **Homebase:** ADHD-optimized household command center with heuristics for anomaly detection, distributed notification loops, and multi-residence bill tracking.
- **Washboard:** Contactless QR car wash booking system currently undergoing a design and architecture overhaul for multi-tenant scalability.

For a complete list of earth science automation tools, civic tech platforms, and algorithmic utilities, please visit my portfolio:
https://ithinkandicode.space

Work Experience

2009–Present **Science Research Specialist**, *Philippine Institute of Volcanology and Seismology (PHIVOLCS)*

- Process GNSS and geophysical data for tectonic monitoring, earthquake research, and fault assessment across the Philippines.
- Design and build headless processing pipelines, distributed task systems, and full-stack applications (e.g., MOVE Faults, EIL Hazard Suite, PHAST) to automate decades of legacy scientific workflows.
- Transitioned technical skills to modern web development, building production applications that apply scientific rigor to software engineering.

2005–2009 **Software Test Engineer**, *Azeus Systems Philippines*

- Performed functional and regression testing for large-scale web applications, including government contract platforms.
- Wrote detailed manual test cases and reported issues with clear reproduction steps, improving dev handoff quality.
- Used SQL for test setup and validation of back-end logic and edge cases during integration testing.

2004–2005 **Science Information Analyst**, *ESS Manufacturing Company, Inc. (EMCI)*

2002–2003 **Student Assistant**, *National Institute of Physics – UP Diliman*

Education

2015 **Master of Geoscience (Partially Completed)**, *Macquarie University*

1999–2004 **BS Applied Physics, Instrumentation**, *University of the Philippines Diliman*
Eugenio Lopez Jr. Scholar (2002–2004)

Select Publications & Research

2023 *Geophysical Research Letters:* Oblique blind faulting, Abra Earthquake

2013 *Journal of Geophysical Research: Solid Earth:* Interseismic deformation along the Philippine Fault

2015 *Journal of Disaster Research:* Plate convergence and GPS motions in Mindanao

2010–2013 *AGU Fall Meeting Abstracts:* Deformation modeling and GPS-based intrusion studies

(Full list available on request or via LinkedIn/Google Scholar profile)